"On my honor, I have neither given nor received any unauthorized and/or inappropiate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues" Irfan Yuceson 150720030

1-

$$A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & B & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

③ $-\frac{1}{2}R_3+R_4\rightarrow R_4$ } Third

① $(-8)\,R_1+R_2\rightarrow R_2$
$(m).R_1+R_3\rightarrow R_3$
$(-1).R_1+R_4\rightarrow R_4$
} First we have to do these

② $(-1)\,R_2+R_3\rightarrow R_3$
$\&R_2+R_4\rightarrow R_4$
} second

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & 8 & 3 & \alpha \\ 1 & 14 & -6 & 2 \end{bmatrix}$$

$(-3)R_1 + R_2 \to R_2$

$mR_1 + R_3 \to R_3$

$(-1)R_1 + R_4 \to R_4$

$$\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha-8 & 5-8\alpha & -9\gamma+8\gamma & 1-8\alpha \\ 2+m & 8+m\alpha & 3 & \alpha^2+m\alpha \\ 0 & 14-\alpha & -6 & 2-\alpha \end{bmatrix}$$

$3\alpha - 8 = 0$

$\alpha = \frac{8}{3}$

$2 + m = 0$

$m = -2$

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -8 & -61/3 \\ 0 & 8 + \frac{(-186)}{3} & 3 & 4/3 \\ 0 & \frac{34}{3} & -6 & \frac{-2}{3} \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & B+(-11/3 & 3 & 4/3 \\ 0 & \frac{34}{3} & -6 & -2/3 \end{bmatrix}$$

$(-1)R_2+R_3 \to R_3$

$\xi R_2+R_4 \to R_4$

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & B+(-11/3)+\frac{49}{3} & 3+\gamma & 65/3 \\ 0 & \frac{34}{3}-\frac{49\xi}{3} & -6+\xi(-\gamma) & -\frac{2}{3}-\frac{61\xi}{3} \end{bmatrix}$$

$B+\frac{33}{3}=0$

$B=-11$

$\frac{34}{3}=\frac{49\xi}{3}=0$

$\xi=\frac{34}{49}$

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & 0 & 3+\gamma & 65/3 \\ 0 & 0 & -6-\frac{34\gamma}{49} & -\frac{2}{3}-\frac{61,34}{3} \end{bmatrix}$$

$-\frac{1}{2}R_3+R_4 \to R_4$

$-\frac{1}{2}(3+\gamma)-6-\frac{34\alpha}{49}=0$

$-\frac{3}{2}-\frac{\gamma}{2}-6-\frac{34\alpha}{49}=0$

$-\frac{15}{2}=\frac{\gamma}{2}+\frac{34\gamma}{49}=\frac{(49+68)\gamma}{98}$

$\frac{-15}{2}=\frac{117}{98}\gamma$

$\boxed{\gamma=\frac{-245}{39}}$

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2- 'ℓ' dimensional space

$x = 5y$ intersection with another geometry gives a volume. (3D)

a) Minimum dimension of space with comments
if we give ℓ and the other geometry 3D. since $x=5y$ is 2D it will lower the dimension of the ℓ to 2D from 3D and if they interset both of them make a line.

b) Propose an expression for the unknown geometry to have a volume in intersection

if we

c) $x+4y-z=6$. Find all vectors in the intersection.

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3- Suppose u is a unit vector in $\mathbb{R}^n$. This problem is about the n by n symmetric matrix $H = I - 2uu^T$

a) $H^2 = I$ Show!

$u^T u = 1$ $\quad H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4uu^T uu^T$

$\Rightarrow H^2 = I$

Symmetric $\Rightarrow H = H^T$; $H \cdot H^T = I$

b) H is orthonormal matrix, Show.

→

c) one eigenvector of H is u itself: eigenvalue?

$Hu = (I - 2uu^T)u = u - 2uu^Tu = u - 2u = -u$

$\lambda = -1$ //

d) Is H diagonalizable? Why?

yes.

if u and v are orthogonal, $Hv = (I - 2uu^T)v = v - 2uu^Tv = v$

Space spanned by u has dimension of n-1. Then there is a basis of (n-1) eigenvectors for this eigenvalue. If we add u, then we can say H is diagonalizable.

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